



ASCHAM SCHOOL

YEAR 12
MATHEMATICS EXTENSION 1
TRIAL EXAMINATION 2012

GENERAL INSTRUCTIONS

5 minutes reading time
working time 2 hours
use black or blue pen
a table of standard integral is provided
approved calculators and templates may be used.

Total Marks - 70

Section 1 – MULTIPLE CHOICE (1 mark each)

- Attempt Questions 1-10
- Allow 15 minutes
- Answers on the sheet provided at the back of exam.
Write your name/NUMBER, teacher's name

Section 2– Question 11 – 14 (15 marks each)

- Allow 1 hour 45 minutes
- Start each question in a new booklet.
- If you use a second booklet for a question, place it inside the first.
- Write your name/NUMBER, teacher's name and question number on each booklet.

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Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section 1 Multiple choice

(1 mark each)

(Mark the correct answer on the sheet provided)

1. A particle moves in a straight line and its position at any time t is given by $x = 3\cos 2t$. The motion is simple harmonic. What is the greatest speed?

(A) 3
(B) -6
(C) 2
(D) 6

2. The number N of animals in a population is increasing. At time t years the population is given by $N = 100 + Ae^{kt}$ for constants $A > 0$ and $k > 0$. Which of the following is the correct differential equation?

(A) $\frac{dN}{dt} = k(N - 100)$
(B) $\frac{dN}{dt} = -k(N + 100)$
(C) $\frac{dN}{dt} = -k(N - 100)$
(D) $\frac{dN}{dt} = k(N + 100)$

3. $\frac{d}{dx}(\sec \frac{x}{2})$ is

(A) $\sec \frac{x}{2} \tan \frac{x}{2}$
(B) $2 \sec \frac{x}{2} \tan \frac{x}{2}$
(C) $\frac{1}{2} \sec \frac{x}{2} \tan \frac{x}{2}$
(D) $\frac{1}{2} \tan^2 \frac{x}{2}$

4. The speed v (m/s) of a particle moving in a straight line is given by $v^2 = 6 + 4x - 2x^2$, where its displacement from a fixed point O is x m. The motion is simple harmonic. What is the centre of the motion?

- (A) $x = -2$
(B) $x = -1$
(C) $x = 1$
(D) $x = 2$

5. What is the domain and range of $y = \cos^{-1}(\frac{3x}{2})$?

- (A) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$. Range: $0 \leq y \leq \pi$
(B) Domain: $-\frac{3}{2} \leq x \leq \frac{3}{2}$. Range: $0 \leq y \leq \pi$
(C) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$. Range: $-\pi \leq y \leq \pi$
(D) Domain: $-\frac{3}{2} \leq x \leq \frac{3}{2}$. Range: $-\pi \leq y \leq \pi$

6. An object is projected with a velocity of 30 ms^{-1} at an angle of $\tan^{-1} \frac{3}{4}$ to the horizontal. What is the initial vertical component of its velocity?

- (A) 18 ms^{-1}
(B) 50 ms^{-1}
(C) $30 \tan \frac{3}{4} \text{ ms}^{-1}$
(D) $30 \sin \frac{3}{4} \text{ ms}^{-1}$

7. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

- (A) $f^{-1}(x) = e^{y-2}$
(B) $f^{-1}(x) = e^{y+2}$
(C) $f^{-1}(x) = \log_e x - 2$
(D) $f^{-1}(x) = \log_e x + 2$

8. Which of the following is the correct expression for $\int \frac{dx}{\sqrt{49-x^2}}$?

(A) $-\cos^{-1} \frac{x}{7} + c$

(B) $\cos^{-1} 7x + c$

(C) $-\sin^{-1} \frac{x}{7} + c$

(D) $\sin^{-1} 7x + c$

9. Given that $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = 0$, the unique value of $\sin^{-1} x + \cos^{-1} x$ is?

(A) C , where C is a constant

(B) 0

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

10. The integral of $\cos^2 2x$ is?

(A) $\frac{\cos^3 2x}{3} + C$

(B) $\left(\frac{\cos 2x}{6}\right)^3 + C$

(C) $\frac{x}{2} + \frac{\sin 4x}{8} + C$

(D) $\frac{x}{2} + \frac{\sin 2x}{4} + C$

Section 2: Answer in booklets provided**Question 11 Begin a new booklet**

- a) Find the product of the roots of the polynomial $P(x)=2x^4-x^3+2x-3$ **(1)**
- b) Find $\frac{d}{dx}(\tan^{-1} 2x^2)$. **(2)**
- c) If P is the point $(-2, 3)$ and K is the point $(10, 1)$ find the coordinates of the point J which divides the interval PK externally in the ratio $4:3$. **(2)**
- d) Find the acute angle between the lines $y=x$ and $\sqrt{3}y=-x$. **(2)**
- e) Solve $\frac{5}{x+2} \leq 1$. **(3)**
- f) Find an expression for $\sin(\tan^{-1} x)$. **(2)**
- g) Solve the equation $\sin 2x=2 \sin^2 x$ for $0 \leq x \leq 2\pi$. **(3)**

Question 13 **Begin a new booklet**

a) Express $\cos x - \sin x$ in the form $A \cos(x + \alpha)$, with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. (2)

b) Hence find the general solution to $\cos x - \sin x = \frac{1}{\sqrt{2}}$. (2)

c) The metal surface of a heater is cooling after it has been switched off. The room has a constant temperature of 20°C . At a time t minutes, its temperature M decreases according to the equation $\frac{dM}{dt} = -k(M - 20)$ where k is a positive constant. The initial temperature of the metal was 160°C and it cools to 100°C after 5 minutes.

i) Show that $M = 20 + Ae^{-kt}$ is a solution to the differential equation, where A is a constant. (1)

ii) Find the values of A and k . (2)

iii) How long will it take for the metal surface to cool to 40°C to the nearest minute. (2)

d) A vitamin tablet which dissolves in water is in the shape of a cylinder. The radius of its cross section is initially 0.5 cm and the height of the tablet is always twice the radius when it dissolves. The tablet dissolves completely after 5 minutes. The rate at which the tablet dissolves is proportional to its surface area A such that $\frac{dV}{dt} = pA$ where p is a constant.

i) Show that $\frac{dr}{dt} = p$ where r is the radius of the tablet in cm at time t minutes. (3)

ii) Find r as a function of t . (2)

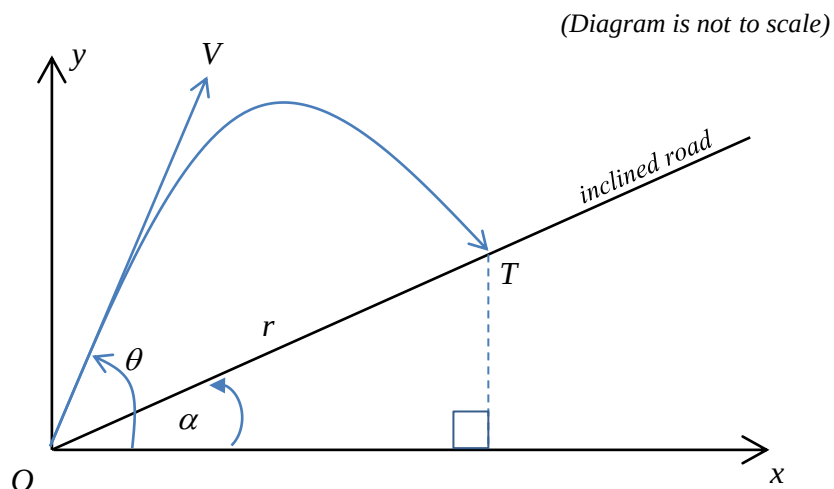
iii) Find the value of p . (1)

Question 14 **Begin a new booklet**

- a) The acceleration of a particle P is given by $\ddot{x} = 8x(x^2 + 4)$ where x is the displacement of P from a fixed point O after t seconds. Initially the particle is at O and has a velocity of 8 m/s in the positive direction.
- i) Show that the speed of the particle is given by $2(x^2 + 4)$ m/s. **(3)**
 - ii) Explain why the velocity of the particle is always positive. **(1)**
 - iii) Hence find the time taken for the particle to travel 2 metres from O . **(3)**
- b) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$. The equation of the tangent at P is given by $y = px - ap^2$ (do not prove this).
- i) Show that the tangents at P and Q meet at T , where T is the point $(a(p+q), apq)$. **(2)**
 - ii) As P moves, Q is chosen such that $\angle POQ$ is always 90° , where O is the origin. Find the locus of T . **(2)**

Question 14 continues on the next page

- c) The diagram shows an inclined road which makes an angle of α with the horizontal.



A projectile is fired from O , at the bottom of the inclined road, with a speed of V m/s at an angle of elevation θ to the horizontal as shown above. Using the axes above, you may assume that the position of the projectile is given by

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{1}{2}gt^2$$

where t is the time, in seconds, after firing, and g is the acceleration due to gravity.

For simplicity, assume that $\frac{2V^2}{g} = 1$.

- i) Show that the path of the trajectory of the projectile is

$$y = x \tan \theta - x^2 \sec^2 \theta . \quad (2)$$
- ii) Show that the range of the projectile $r = OT$ metres, up the inclined road is given by

$$r = \frac{\sin(\theta - \alpha) \cos \theta}{\cos^2 \alpha} . \quad (2)$$

Student Number: _____

Name: _____

SECTION I **Extension 1 Multiple Choice Answer Sheet****10 Marks****This sheet must be detached and handed in separately****Shade the correct answer:**

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

SECTION 1:

1) $x = 3 \cos 2t$
 $\dot{x} = -6 \sin 2t$

∴ greatest speed = 6

2) (A)

3) $\frac{d}{dx} \left(\sec \frac{x}{2} \right)$
 $= \frac{1}{2} \sec \frac{x}{2} \tan \frac{x}{2}$

4) Centre when acc. is zero
 $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2 - 2x$
 $0 = 2 - 2x$
 $x = 1$

5) range: $0 \leq y \leq \pi$
 domain: $-1 \leq \frac{3x}{2} \leq 1$

$-2 \leq 3x \leq 2$
 $-\frac{2}{3} \leq x \leq \frac{2}{3}$

(A)

6) Vertical component is
 $v \sin \theta = 30 \times \frac{3}{5}$
 $= 18 \text{ m/s.}$ (A)



7) $y = e^{x+2}$

$x = e^{y+2}$
 $\ln x = y + 2$
 $y = \ln x - 2$ (C)

8) (A)

9) (D)

10) $\cos^2 2x = \frac{1}{2} + \frac{1}{2} \cos 4x$
 $\therefore \int \frac{1}{2} + \frac{1}{2} \cos 4x \, dx$
 $= \frac{x}{2} + \frac{\sin 4x}{8} + C$

(C)

DAACA
 ACADC

QUESTION 11:

a) $\beta x = \frac{Q}{a}$
 $= -\frac{3}{2}$

c) P(-2, 3) & K(10, 1)
 externally in ratio 4:3
 Let J = (x, y)

$x = \frac{-3(-2) + 4(10)}{4-3}$
 $= 46$

$y = \frac{-3(3) + 4(1)}{4-3}$
 $= -5$

∴ J = (46, -5)

e) $\frac{5}{x+2} \leq 1 \dots x \neq -2$

$\frac{5}{x+2} \times (x+2)^2 \leq 1 \times (x+2)^2 \dots x \neq -2$

$5(x+2) \leq (x+2)^2 \dots x \neq -2$

$(x+2)^2 - 5(x+2) > 0 \dots x \neq -2$

$(x+2)(x+2-5) > 0 \dots x \neq -2$

$(x+2)(x-3) > 0 \dots x \neq -2$

∴ $x < -2$ OR $x > 3 \dots x \neq -2$

(b) $\frac{d}{dx} (\tan^{-1} 2x^2)$

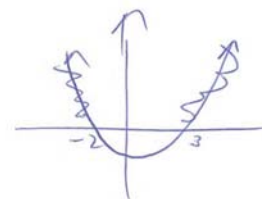
$= \frac{1}{1+(2x^2)^2} \times 4x$
 $= \frac{4x}{1+4x^4}$

d) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$\tan \theta = \left| \frac{1 - \frac{1}{\sqrt{3}}}{1 + (1) \times (\frac{1}{\sqrt{3}})} \right|$

$\tan \theta = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$

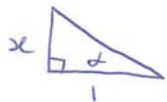
$\theta = 75^\circ$



(7) $\sin(\tan^{-1}x)$

let $\tan^{-1}x = \theta$

$\therefore x = \tan \theta$



using pythagoras,
hypotenuse = $\sqrt{x^2+1}$

$\therefore \sin \theta = \frac{x}{\sqrt{x^2+1}}$

$\therefore \sin(\tan^{-1}x) = \frac{x}{\sqrt{x^2+1}}$

8) $\sin 2x = 2\sin^2 x$

$2\sin x \cos x = 2\sin^2 x$

$2\sin^2 x - 2\sin x \cos x = 0$

$2\sin x (\sin x - \cos x) = 0$

$\therefore 2\sin x = 0$ OR $\sin x = \cos x$

$\sin x = 0$ OR $\tan x = 1$

$\therefore x = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$

QUESTION 12:

a) $\angle XYZ = 90^\circ$ (angle in semi circle)

$\angle ARB = 90^\circ$ (exterior $\angle$ of cyclic quad.)

b) $\int_0^2 \frac{4x}{\sqrt{4x+1}} dx$

$= \frac{1}{4} \int \frac{u-1}{\sqrt{u}} du$

$= \frac{1}{4} \int_1^9 u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$

$= \frac{1}{4} \left[\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right]_1^9$

$= \frac{1}{4} \left[\left(\frac{2}{3} (9)^{\frac{3}{2}} - 2(9)^{\frac{1}{2}} \right) - \left(\frac{2}{3} (1)^{\frac{3}{2}} - 2(1)^{\frac{1}{2}} \right) \right]$

$= \frac{1}{4} \left[(18 - 6) - \left(\frac{2}{3} - 2 \right) \right]$

$= \frac{1}{4} \left[12 + \frac{4}{3} \right] = \frac{10}{3}$

c) $f(x) = e^x - 3x$ & $f'(x) = e^x - 3$

$f(1) = e - 3$

$= -0.2817 \dots < 0$

$f(1.7) = e^{1.7} - 3 \times 1.7$

$= 0.3739 \dots > 0$

change of sign between

$x=1$ & $x=1.7$

$\therefore$ root lies between

$x = \frac{1}{4}(u-1)$

$4x = u-1$

$u = 4x+1$

$\boxed{du = 4dx}$

$x=0: 0 = \frac{1}{4}(u-1)$

$u=1$

$x=2: 2 = \frac{1}{4}(u-1)$

$u=9$

$x_1 = 1.7 - \frac{e^{1.7} - 3(1.7)}{e^{1.7} - 3}$

$x_1 = 1.54$ (approx)

$\therefore x_1 \approx 1.55$ (3 s.f.)

$\therefore$ an approx. to the root is $x \approx 1.55$.

$$d) p(x) = x^3 + ax^2 + bx + c$$

$$p(0) = 0 = c$$

$$\therefore \boxed{c=0}$$

$$p(3) = 0 = 27 + 9a + 3b \quad (1)$$

$$p(-3) = 0 = -27 + 9a - 3b \quad (2)$$

$$1) a = 0$$

$$\boxed{a=0}$$

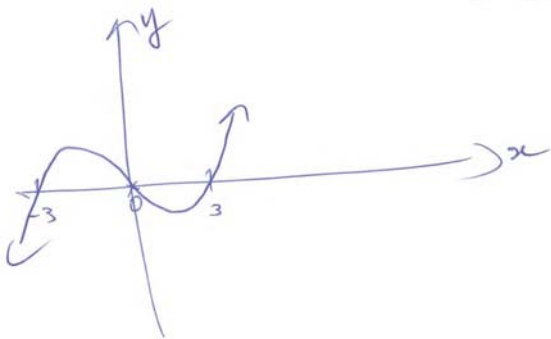
$$\text{Subst } a=0 \text{ into (1): } 27 + 0 + 3b = 0$$

$$\boxed{b = -9}$$

$$\therefore p(x) = x^3 - 9x^2$$

$$p(x) = x(x^2 - 9)$$

$$= x(x-3)(x+3)$$



$$e) \therefore 5^n - 1$$

$$\text{Prove for } n=1: 5^1 - 1 = 4$$

$$\therefore \text{true for } n=1$$

$$\text{Assume true for } n=k: \text{ where } k \in \mathbb{N}$$

$$5^k - 1 \text{ is divisible by 4}$$

$$\therefore \text{Let } 5^k - 1 = 4P \text{ where } P \in \mathbb{N}$$

$$\text{Prove for } n=k+1:$$

$$5^{k+1} - 1 = 5^k \times 5 - 1$$

$$= (4P+1) \times 5 - 1$$

$$= 20P + 5 - 1$$

$$= 20P + 4$$

$$= 4(5P+1)$$

$$= 4M \dots \text{ where } M = 5P+1$$

$$\therefore 5^{k+1} - 1 \text{ is divisible by 4}$$

QUESTION 13:

a) $\cos x - \sin x = A \cos x \cos t - A \sin x \sin t$

Equating: $A \cos t = 1$ -- (1)

$A \sin t = 1$ -- (2)

Squaring & adding (1) & (2): $A^2 \cos^2 t + A^2 \sin^2 t = 2$

$A^2 (\cos^2 t + \sin^2 t) = 2$

$A^2 = 2$

$A = \sqrt{2}$

From (1) & (2): $\tan t = 1$

$t = \frac{\pi}{4}$... only 1st quad since $\cos t$ & $\sin t$ both > 0

$\therefore \cos x - \sin x = \sqrt{2} \cos(x + \frac{\pi}{4})$

b) $\cos x - \sin x = \frac{1}{\sqrt{2}}$

$\sqrt{2} \cos(x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$

$\cos(x + \frac{\pi}{4}) = \frac{1}{2}$

$x + \frac{\pi}{4} = \frac{\pi}{3} + 2n\pi$ OR $x + \frac{\pi}{4} = -\frac{\pi}{3} + 2n\pi$

$x = \frac{\pi}{12} + 2n\pi$ OR $x = -\frac{7\pi}{12} + 2n\pi$

$\therefore$ general sol'n is

$x = \frac{\pi}{12} + 2n\pi$ OR $x = -\frac{7\pi}{12} + 2n\pi$

(7)

c) $\frac{dM}{dt} = -k(M-20)$

LHS = $A e^{-kt} \times -k$

$= -kA e^{-kt}$

RHS = $-k(20 + A e^{-kt} - 20)$

$= -kA e^{-kt}$

$= \text{LHS}$

ii) When $t=0$, $M=160$

$160 = 20 + A e^0$

$A = 140$

$\therefore M = 20 + 140 e^{-kt}$

When $t=5$, $M=100$

$100 = 20 + 140 e^{-5k}$

$\frac{80}{140} = e^{-5k}$

$\ln(\frac{4}{7}) = -5k$

$\therefore k = \frac{\ln(\frac{4}{7})}{-5} (\approx 0.11192)$

(iii) $M=40$, $t=?$

$M = 20 + 140 e^{-kt}$

$40 = 20 + 140 e^{-kt}$

$\frac{20}{140} = e^{-kt}$

$\uparrow \ln(\frac{1}{7}) = -kt$
 $\frac{-\ln 7}{-k} = t$

$t = 17.3866...$

$t \approx 17 \text{ min.}$

(8)

d)  $h = 2r$

$$V = \pi r^2 h$$

$$V = \pi r^2 \times 2r$$

$$V = 2\pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$\boxed{\frac{dV}{dt} = 6\pi r^2 \times \frac{dr}{dt}}$$

$$\frac{dV}{dt} = pA$$

$$= p(2\pi r^2 + 2\pi rh)$$

$$= p(2\pi r^2 + 2\pi r \times 2r)$$

$$\frac{dV}{dt} = p \times 6\pi r^2$$

$$\therefore 6\pi r^2 \times \frac{dr}{dt} = p \times 6\pi r^2$$

$$\frac{dr}{dt} = p$$

$$\text{ii) } r = \int \frac{dr}{dt}$$

$$= \int p \, dt$$

$$\therefore r = pt + C$$

$$\begin{aligned} t=0, r=0.5 \\ \therefore 0.5 = p \times 0 + C \\ C = 0.5 \\ \therefore \boxed{r = pt + 0.5} \end{aligned}$$

$$\begin{aligned} \text{ii) } 0, t=5 \\ 0 = p \times 5 + 0.5 \\ \underline{\underline{p = -0.1}} \end{aligned}$$

QUESTION 14

$$\text{a) i) } \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 8x(x^2 + 4)$$

$$\begin{aligned} \frac{1}{2} v^2 &= \int 8x^3 + 32x \, dx \\ \frac{1}{2} v^2 &= 2x^4 + 16x^2 + C \\ v^2 &= 4x^4 + 32x^2 + D \end{aligned}$$

when $x=0$, $v=8$:

$$64 = 0 + 0 + D$$

$$D = 64$$

$$\begin{aligned} \therefore v^2 &= 4x^4 + 32x^2 + 64 \\ &= 4(x^4 + 8x^2 + 16) \end{aligned}$$

$$v^2 = 4(x^2 + 4)$$

$$\therefore v = \pm 2(x^2 + 4)$$

$$\therefore \text{speed} = |\pm 2(x^2 + 4)|$$

$$s = 2(x^2 + 4) \, \text{m/s}$$

ii) velocity always +ve since it is initially 8 m/s & v needs to be zero before it can become -ve, but $2(x^2 + 4) > 0$ for all x .

$$\begin{aligned}
 \text{ii)} \quad v &= 2(x^2+4) \\
 \frac{dx}{dt} &= 2(x^2+4) \\
 \frac{dt}{dx} &= \frac{1}{2(x^2+4)} \\
 t &= \frac{1}{2} \int \frac{1}{x^2+4} dx \\
 t &= \frac{1}{2} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + C \\
 t &= \frac{1}{4} \tan^{-1} \frac{x}{2} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{when } t=0, x=0: \\
 0 &= \frac{1}{4} \tan^{-1} 0 + C \\
 \therefore C &= 0
 \end{aligned}$$

$$\therefore t = \frac{1}{4} \tan^{-1} \left(\frac{x}{2} \right)$$

$$\begin{aligned}
 \text{when } x=2: \quad t &= \frac{1}{4} \tan^{-1} \left(\frac{2}{2} \right) \\
 t &= \frac{1}{4} \tan^{-1} 1 \\
 t &= \frac{1}{4} \times \frac{\pi}{4} \\
 t &= \frac{\pi}{16} \text{ sec.}
 \end{aligned}$$

QUESTION 4:
a) Eq'n of tangents at P & Q respectively
 $y = px - ap^2$ and $y = qx - aq^2$

$$\begin{aligned}
 \text{Equating:} \quad px - ap^2 &= qx - aq^2 \\
 x(p-q) &= a(p^2 - q^2) \\
 x(p-q) &= a(p-q)(p+q) \\
 \therefore x &= a(p+q)
 \end{aligned}$$

$$\begin{aligned}
 \text{Subst in } y &= px - ap^2 \\
 y &= p \times a(p+q) - ap^2 \\
 y &= ap^2 + apq - ap^2 \\
 y &= apq \\
 \therefore T &= (a(p+q), apq)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } C_{\text{red op}} \times C_{\text{red oa}} &= -1 \\
 \frac{ap^2}{2dp} \times \frac{aq^2}{2dq} &= -1 \\
 \frac{pq}{4} &= -1 \\
 \boxed{pq} &= -4
 \end{aligned}$$

$$\text{Locus of } T: \quad x = a(p+q) \quad \& \quad y = apq$$

$$\begin{aligned}
 \text{Subst } pq = -4 \text{ into } y &= apq \\
 y &= a \times -4 \\
 \boxed{y = -4a} &\therefore \text{Locus of } T \quad (12)
 \end{aligned}$$

$$\therefore \text{c) i) } x = Vt \cos \theta \quad \text{--- ①}$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2 \quad \text{--- ②}$$

$$\text{from ① } t = \frac{x}{V \cos \theta}$$

Subst in ②:

$$y = V \times \frac{x}{V \cos \theta} \times \sin \theta - \frac{1}{2}g \times \frac{x^2}{V^2 \cos^2 \theta}$$

$$= x \tan \theta - x^2 \sec^2 \theta \times \frac{g}{2V^2}$$

$$= x \tan \theta - x^2 \sec^2 \theta \times 1 \quad \because \text{since } \frac{2V^2}{g} = 1$$

$$y = x \tan \theta - x^2 \sec^2 \theta$$

$$\text{ii) Let } T = (x, y)$$

$$\text{then } x = r \cos \theta$$

$$\text{and } y = r \sin \theta$$

$$\text{From i) : } y = x \tan \theta - x^2 \sec^2 \theta$$

$$r \sin \theta = r \cos \theta \times \tan \theta - r^2 \cos^2 \theta \times \sec^2 \theta$$

$$r \cos^2 \theta \sec^2 \theta = \cos \theta \tan \theta - \sin \theta$$

$$r = \frac{\cos \theta \tan \theta - \sin \theta}{\cos^2 \theta \sec^2 \theta}$$

$$= \frac{\cos \theta \times \frac{\sin \theta}{\cos \theta} - \sin \theta}{\cos^2 \theta}$$

$$= \frac{\frac{\cos \theta \sin \theta - \sin \theta \cos \theta}{\cos \theta}}{\cos^2 \theta}$$

$$= \frac{\cos \theta \sin \theta - \sin \theta \cos \theta}{\cancel{\cos \theta}} \times \frac{\cos^2 \theta}{\cos^2 \theta}$$

$$r = \frac{\sin(\theta - \theta) \times \cos \theta}{\cos^2 \theta}$$